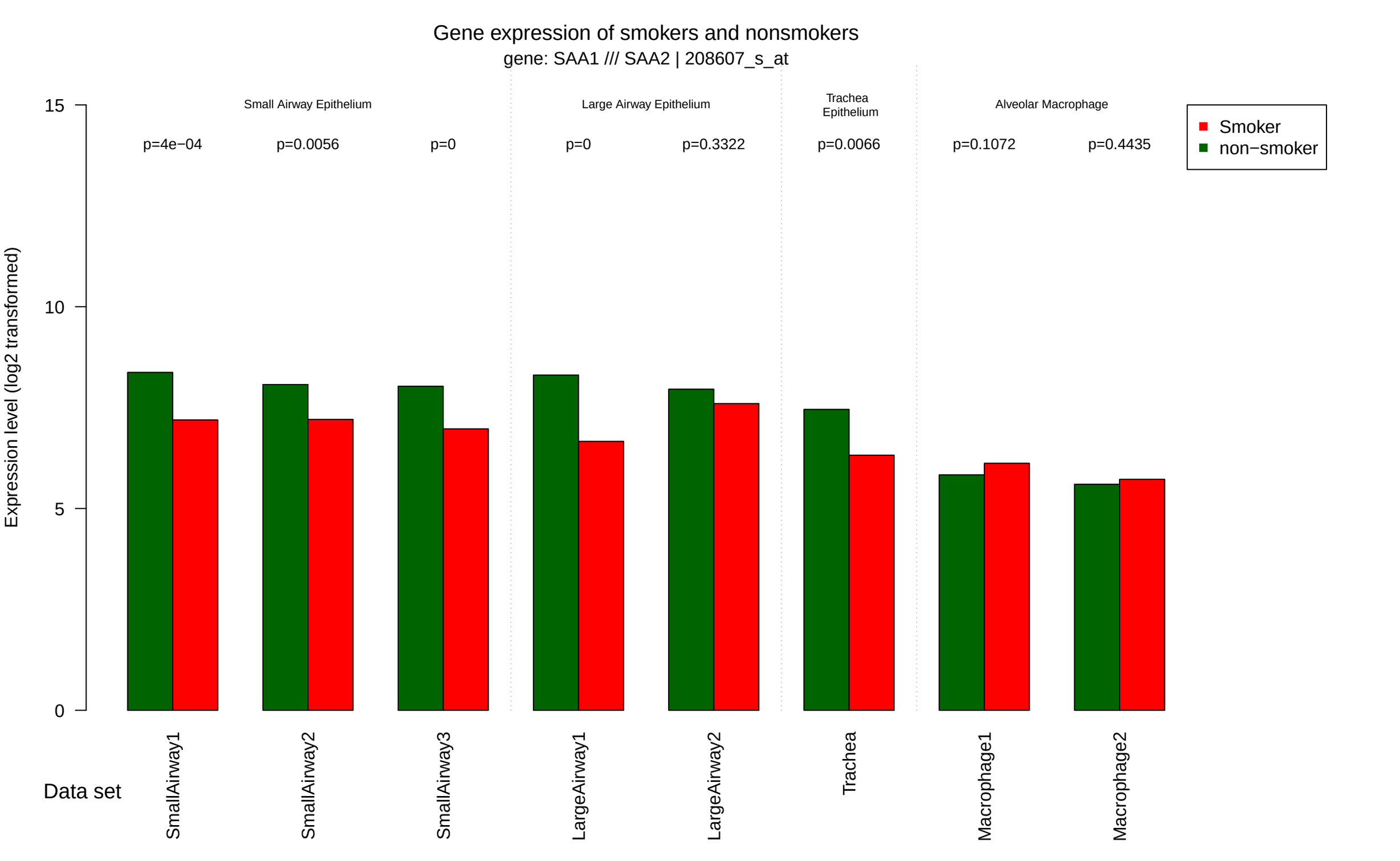
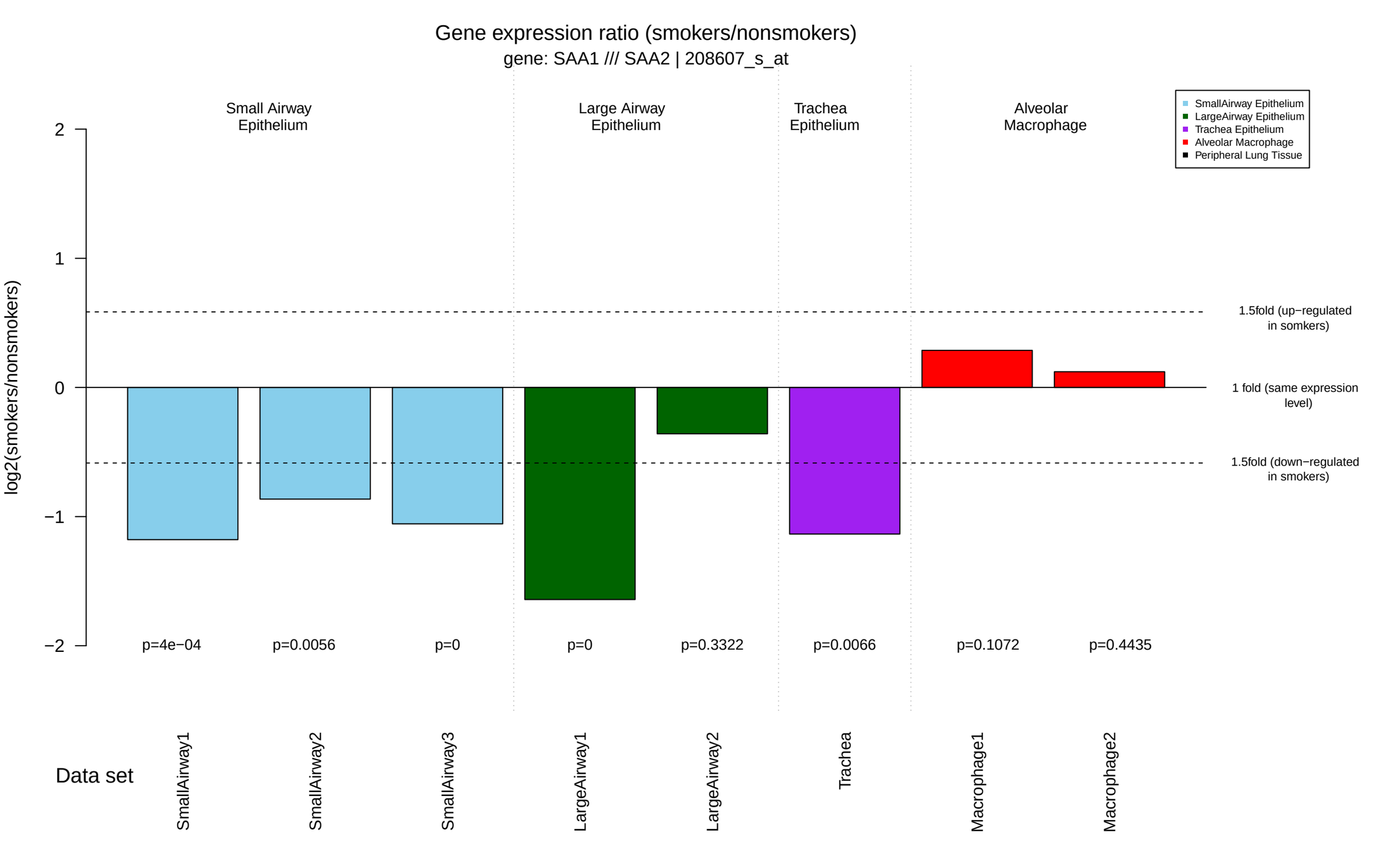
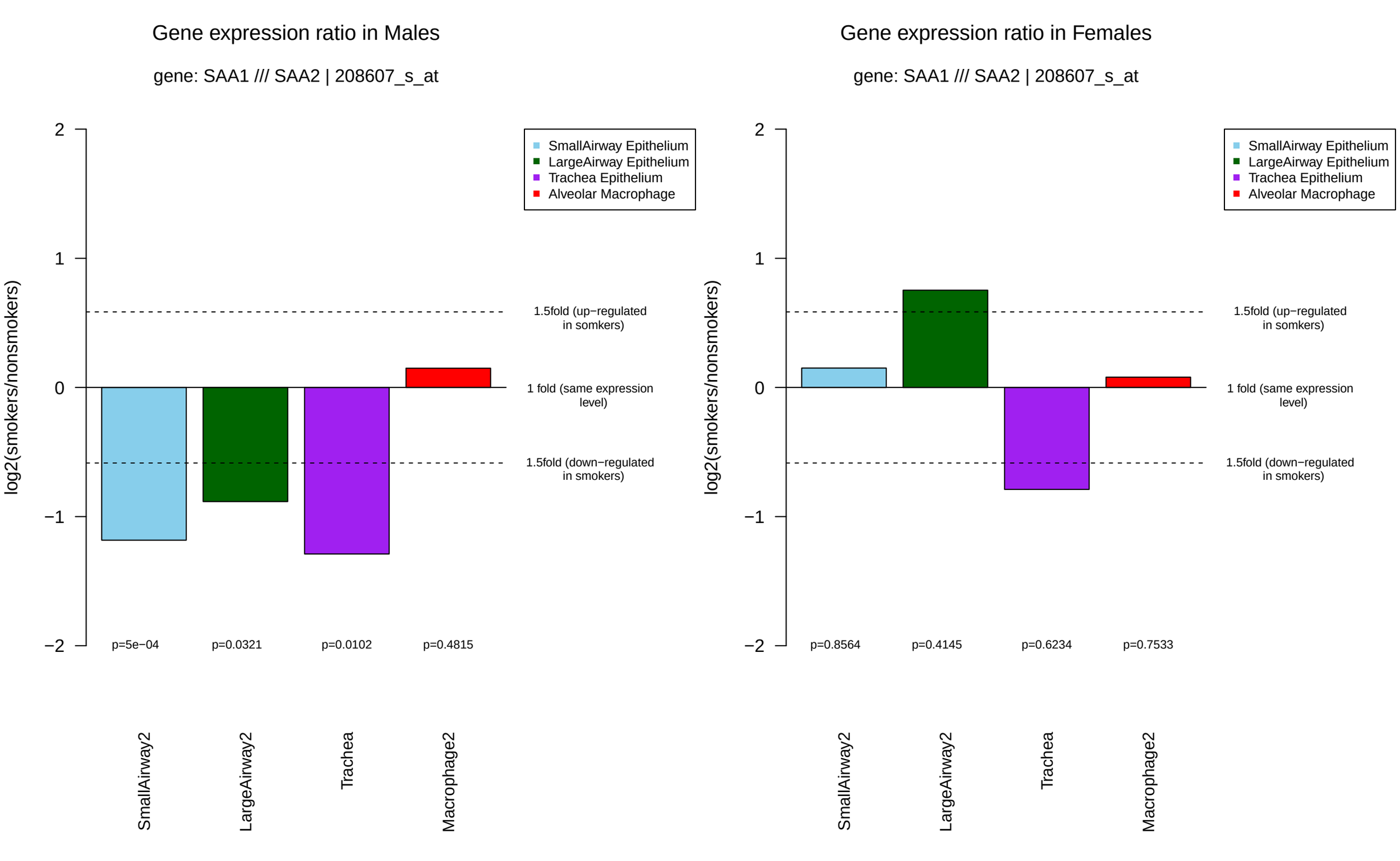
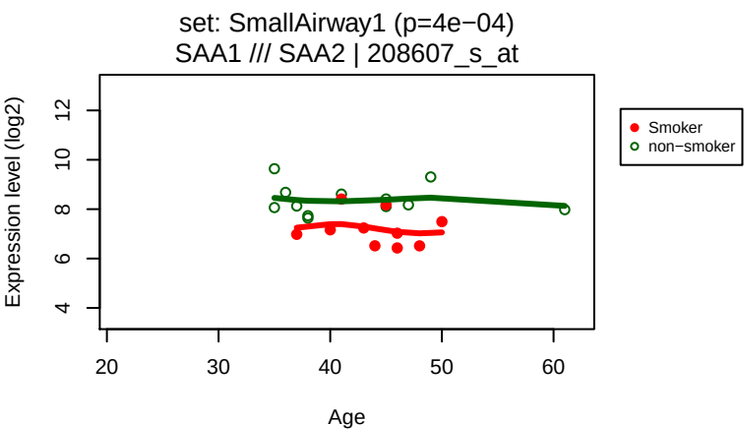
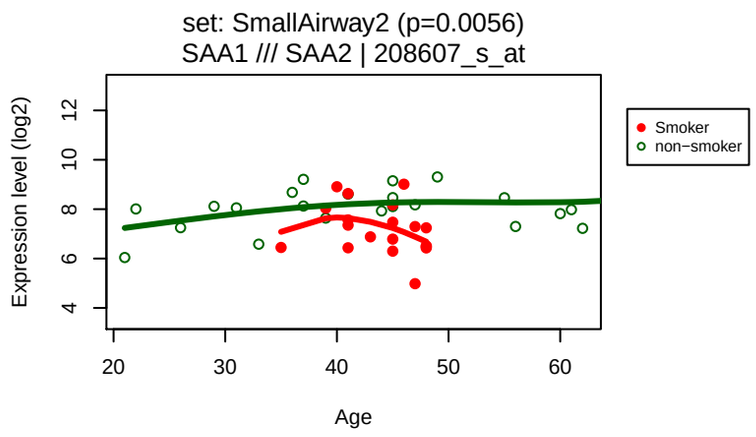
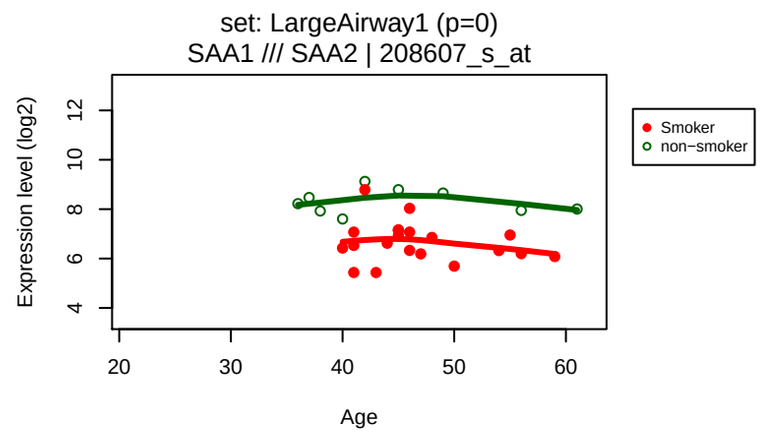

Gene expression of smokers and nonsmokers
gene: SAA1 /// SAA2 | 208607_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=4e−04
p=0.0056
p=0
p=0
p=0.3322
p=0.0066
p=0.1072
p=0.4435
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

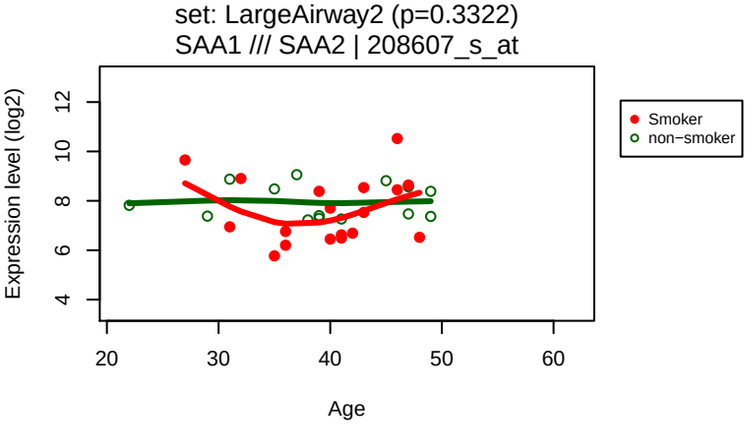

Gene expression ratio (smokers/nonsmokers)
gene: SAA1 /// SAA2 | 208607_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=4e−04
p=0.0056
p=0
p=0
p=0.3322
p=0.0066
p=0.1072
p=0.4435
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

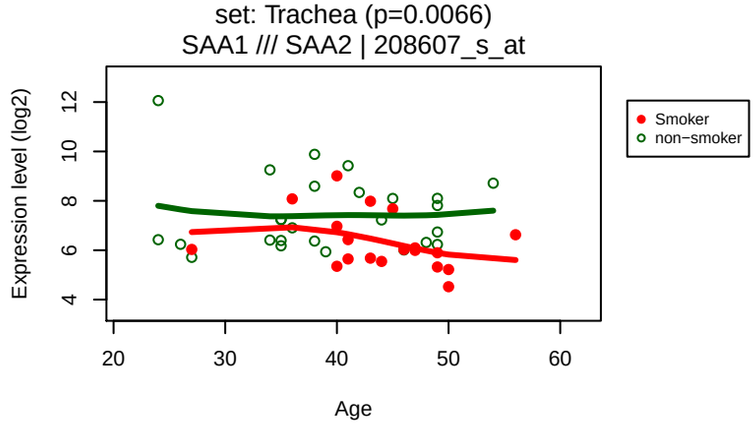

Gene expression ratio in Males
gene: SAA1 /// SAA2 | 208607_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=5e−04
p=0.0321
p=0.0102
p=0.4815
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAA1 /// SAA2 | 208607_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8564
p=0.4145
p=0.6234
p=0.7533
SmallAirway2
LargeAirway2
Trachea
Macrophage2

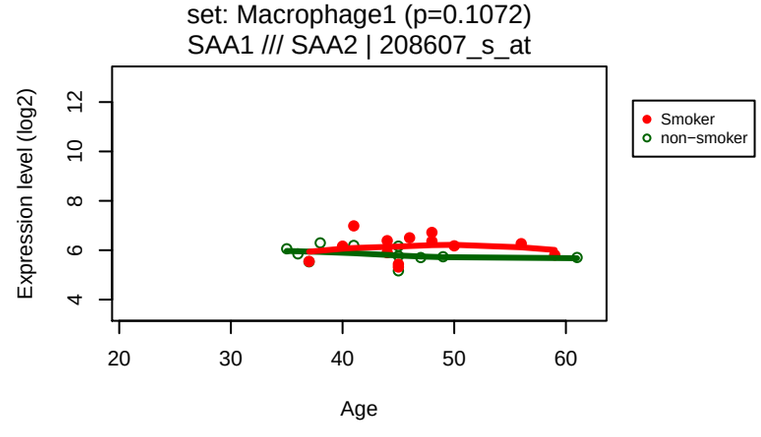

set: SmallAirway1 (p=4e−04)
SAA1 /// SAA2 | 208607_s_at
Expression level (log2)
Age
Smoker
non−smoker

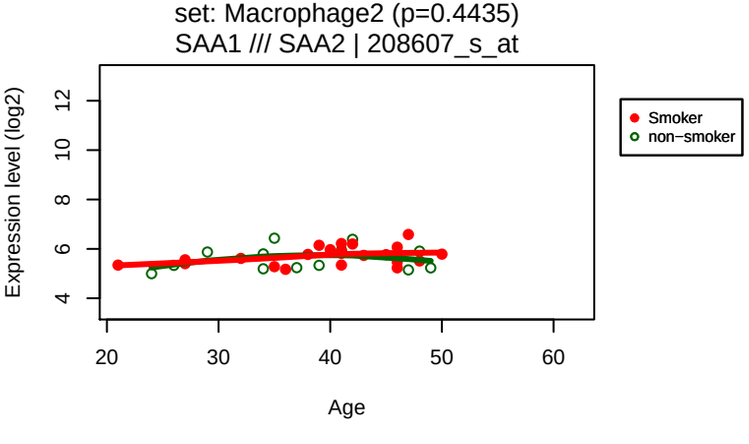

set: SmallAirway2 (p=0.0056)
SAA1 /// SAA2 | 208607_s_at
Expression level (log2)
Age
Smoker
non−smoker

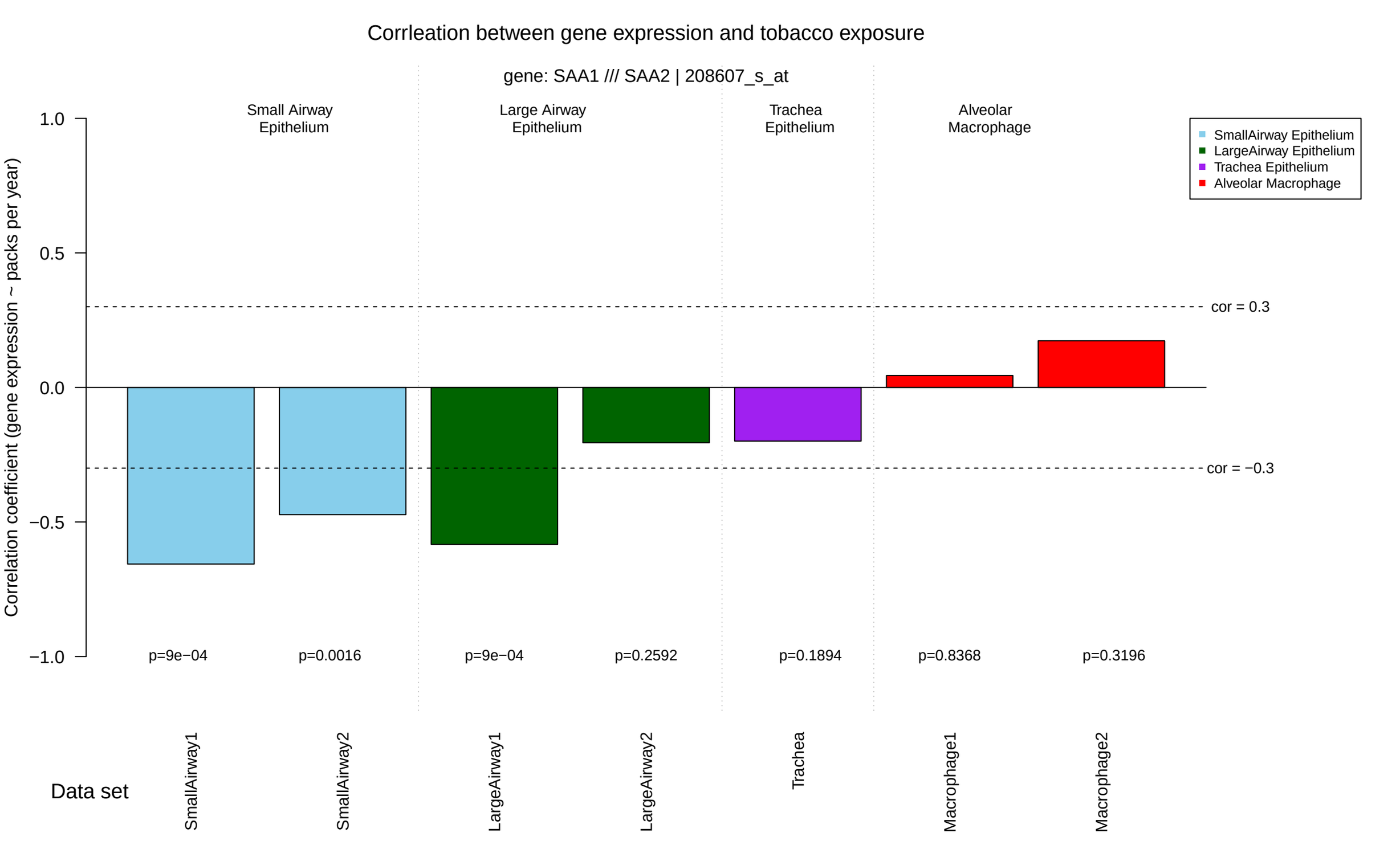

set: LargeAirway1 (p=0)
SAA1 /// SAA2 | 208607_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3322)
SAA1 /// SAA2 | 208607_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0066)
SAA1 /// SAA2 | 208607_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1072)
SAA1 /// SAA2 | 208607_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4435)
SAA1 /// SAA2 | 208607_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAA1 /// SAA2 | 208607_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=9e−04
p=0.0016
p=9e−04
p=0.2592
p=0.1894
p=0.8368
p=0.3196
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2